FIRMWARE MODULE INTEGRATION GUIDELINES

1. Document Information

Field	Description
Project	
Module Name	
Module Version	
Package Version	
Document Version	
Release Date	
Author	
Owner Team	
Target Team	

2. Purpose & Scope

2.1 Purpose

Describe the purpose of the module and explain how another Firmware team can integrate it into an existing firmware project.

2.2 Scope

- Supported functionalities:
- Supported platforms:
- Supported compilers:
- Supported integration environments:
- Out-of-scope functionalities:

3. Module Overview

3.1 Description

Provide a brief description of the module and its main responsibilities.

3.2 Software Architecture

Describe the module architecture and its interactions with other software components.

3.3 Dependencies

Dependency	Version	Mandatory	Description
CMSIS		Yes / No	
HAL / LL		Yes / No	
RTOS		Yes / No	
Other modules		Yes / No	

4. Package Contents

File / Directory	Description	Required
include/Module.h	Public API	Yes
source/Module.c	Module implementation	Yes
config/Module_Config.h	Configuration file	Yes
docs/	Technical documentation	No
templates/	Optional source templates	No

5. Integration Requirements

5.1 Hardware Requirements

Parameter	Requirement
MCU Family	
MCU Reference	
Required Peripherals	
Clock Requirements	
GPIO Requirements	
DMA Requirements	

5.2 Software Requirements

Parameter	Requirement
IDE	
Compiler	
Compiler Version	
Language Standard	C99 / C11
CMSIS Version	
HAL / LL Version	
RTOS	

6. Integration Procedure

6.1 Add Source Files

Describe which source files must be added to the firmware project.

6.2 Configure Include Paths

List the directories that must be added to the compiler include paths.

6.3 Configure Compiler Definitions

Definition	Value	Description
MODULE_DEBUG	0 / 1	Debug mode
MODULE_ENABLED	0 / 1	Module activation

6.4 Module Configuration

Describe the available configuration options in Module_Config.h.

Configuration	Description	Default	Allowed Values
MODULE_MEMORY_ALLOCATION	Memory allocation strategy	Static	0: Static / 1: Dynamic
MODULE_BUFFER_SIZE	Buffer size		

6.5 Linker Configuration

Describe any required linker modifications.

Memory Region	Start Address	Size	Description
ROM			Code and constants
RAM			Runtime data
Heap			Dynamic allocation
Stack			Execution stack

6.6 Initialization Sequence

Describe the initialization order and dependencies.

Example:

1. System initialization.
2. Clock configuration.

3. Peripheral initialization.
4. Module configuration.
5. Module initialization.
6. Module execution.

7. API Description

List the public functions exposed by the module.

API	Description	Parameters	Return
Module_Init()	Initialize module	Configuration	Status
Module_Process()	Execute module processing	None	Status
Module_DeInit()	Deinitialize module	None	Status

7.1 Usage Example

Provide a minimal example showing how to initialize and use the module.

“‘c #include “Module.h” #include “Module_Config.h”

int main(void) { System_Init();

```
Module_Init();

while (1)
{
    Module_Process();
}
```